

Daimler Truck Connect - Predictive Intelligence

Commercial Q&A & Costing Annex

Trained on ~500 vehicles / vertical (transformer) - deployed on ~100,000 vehicles in Truck Connect

Prepared by ByteEdge - 2026-06-26 - FX Rs 94/USD - pricing snapshot 24-Jun-2026 (Azure Central India)

Customer-facing answers are in Sections 1-8. Annex A shows the compute as (hours x per-hour rate); Annex B holds internal commercial notes (cost-to-price bridge, margin %) and is NOT for external distribution. Planning estimates for decision-making, not a quotation.

At a glance

Production compute by deployment size (daily batch scoring, all 3 verticals):

Fleet (once-daily batch)	Compute / month	Compute / year	Per vehicle / year
25,000 vehicles	~Rs 0.55-0.85 L	~Rs 6.5-10.3 L	~Rs 26-41
50,000 vehicles	~Rs 0.8-1.3 L	~Rs 9.8-15.3 L	~Rs 20-31
75,000 vehicles	~Rs 1.0-1.6 L	~Rs 12.4-19.3 L	~Rs 16-26
100,000 vehicles (target)	~Rs 1.2-1.9 L	~Rs 14.6-22.8 L	~Rs 15-23

Twice-daily scoring ~ 1.5-1.8x the monthly figure. Per-vehicle cost falls as the fleet grows (shared platform spread over more trucks).

Model tier - changes TRAINING only, not the run-rate above:

Metric	Small models (small data)	Transformer (~500 veh/vertical)
Model family	Ridge / XGBoost / IForest + small LSTM/TCN	Temporal Fusion Transformer
Training GPU	A10 (24 GB)	A100 (80 GB)
Build / industrialization compute	~Rs 0.7-1.2 L	~Rs 1.3-5.0 L (~ Rs 4-4.5 L envelope)
Production run-rate & per-vehicle	as table above	identical (inference is model-agnostic)
Quarterly retrain, 3 verticals	~Rs 450-1,300 / yr (A10)	~Rs 5,000-11,500 / yr (A100)

Bottom line: at 100,000 vehicles, once-daily scoring costs ~Rs 1.2-1.9 Lakh/month (~Rs 15-23 per vehicle/year). The move to a transformer raises only TRAINING / retraining compute (still a few thousand rupees/year); the production run-rate is unchanged because daily batch inference of a still-small model is cheap (< 15 min/pass on CPU in the consolidated plan).

Contents

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1. How to read this document

- **Customer answers (Sec. 1-8):** clean, decision-ready.
- **Annex A:** compute as hours x per-hour rate, plus the storage / platform lines that are not hour-priced.
- **Annex B (internal):** cost-to-price bridge, margin %, points to confirm. Remove before sharing externally.

Currency Rs 94/USD. Cloud prices: Azure Central India, 24-Jun-2026. Deployment ~100,000 vehicles; models trained on ~500 vehicles/vertical (transformer tier).

2. Programme & Scope

Q1. What does the Rs 24 Lakhs “Clutch 500 Readiness” programme cover?

A. It takes the proven clutch model from pilot to a production-grade capability ready for the fleet and integrated into Daimler Truck Connect, across three workstreams:

- **Data Engineering & Foundation** - production-grade ingestion, cleaning and sentinel-handling, a curated data store and feature store, plus data-quality and silent-telemetry monitors.
- **Predictive Intelligence Industrialization** - hardening the clutch model (anomaly baseline + classifier + RUL with conformal intervals), the champion/challenger retraining loop, and validation gates.
- **Truck Connect Readiness & Architecture** - the API contract, webhook alerts, the signed/licensed container, and the deployment architecture.

The same foundation and Truck Connect plumbing are reused by the alternator and starter verticals - which is why each industrializes for an incremental Rs 7 Lakhs. Clutch is the most mature (~63-day failure lead).

Q2. What is the breakup of the Rs 24 Lakhs?

Workstream	Amount	What it includes
Data Engineering & Foundation	Rs 7 L	Ingestion, cleaning, sentinel rules, curated + feature store, data-quality monitors
Predictive Intelligence Industrialization	Rs 13 L	Clutch model hardening, calibration, champion/challenger loop, LOVO validation, registry
Truck Connect Readiness & Architecture	Rs 4 L	API contract, webhook alerts, signed licensed container, deployment design
Total	Rs 24 L	~30% data / 54% predictive-model industrialization / 16% Truck Connect readiness

Q3. How much compute cost is assumed (build phase)?

A. ~Rs 4-4.5 Lakhs across development, training, storage and validation. For small models the actual build compute is only ~Rs 0.7-1.2 L (the envelope is conservative); for the transformer tier (~500 veh/vertical, A100, ~600 GPU-hours incl. hyper-parameter search) it is ~Rs 1.3-5.0 L - i.e. the Rs 4-4.5 L envelope still covers the typical build. Most of the Rs 24 L is engineering effort, not compute.

Q4. Does the Rs 24 Lakhs cover all tractor-trailer variants?

A. Yes - assuming a similar clutch architecture. Variants sharing the same clutch hardware and signal set are covered by the same model with minor re-calibration. Materially different hardware, torque ratings or duty cycles may need incremental calibration - a small add-on, not a fresh programme. A variant-to-clutch-hardware mapping from Daimler confirms which variants are in scope at no extra cost (Section 8).

3. Deployment Compute (100,000 vehicles, by 25k slab)

Batch-first architecture: a scheduled CPU batch scores the fleet daily; the GPU is summoned only for quarterly retraining. Compute is model-agnostic (small models or transformer score equally cheaply in batch). Hours x rate in Annex A.

Q5. Production compute by deployment slab (once daily)?

Fleet (once-daily batch)	Compute / month	Compute / year	Per vehicle / year
25,000 vehicles	~Rs 0.55-0.85 L	~Rs 6.5-10.3 L	~Rs 26-41
50,000 vehicles	~Rs 0.8-1.3 L	~Rs 9.8-15.3 L	~Rs 20-31
75,000 vehicles	~Rs 1.0-1.6 L	~Rs 12.4-19.3 L	~Rs 16-26
100,000 vehicles (target)	~Rs 1.2-1.9 L	~Rs 14.6-22.8 L	~Rs 15-23

Compute scales sub-linearly with fleet size (storage and per-vehicle CPU grow ~linearly; the platform / monitoring layer is largely fixed), so per-vehicle cost falls from ~Rs 26-41 at 25k to ~Rs 15-23 at 100k.

Q6. Production compute if scoring runs twice daily?

A. ~1.5-1.8x the once-daily figure (e.g. ~Rs 1.8-3.4 Lakh/month at 100k). It does not double: telemetry is ingested and stored once; only the second scoring pass, the extra prediction-history writes and pipeline runs add cost. The operating rhythm (Section 4) is once-daily, so once-daily is the planning figure.

Q7. Per-vehicle compute cost, split by vertical (at 100k)?

A. All three verticals together cost ~Rs 15-23/vehicle/year. The split is roughly even because most of the cost is shared infrastructure (storage, feature pipeline, platform); the model inference itself is negligible:

Vertical	Daily output	Compute (Rs / veh / yr @ 100k)
Clutch	RUL + risk zone (anomaly-based, ~63-day lead)	~Rs 7-9
Alternator	Risk zone + maintenance window + GED=2 alarm	~Rs 5-7
Starter	Risk zone + window + persistent-RED dwell pager	~Rs 5-7
All three (shared)	combined daily card per vehicle	~Rs 15-23

Clutch is marginally higher (anomaly + RUL). Adding a vertical is incremental (~Rs 5-7/veh/yr), not a third of the cost. This is compute cost, distinct from the SaaS price in Section 7.

4. Operating Rhythm

One cadence runs the deployed system; retraining sits at the bottom of it, gated by new labels.

Cadence	Activity	Output	Honest note
Daily	Batch inference	Risk tier + window + alarms (RUL for clutch)	per-component split below
Weekly	Aggregate daily scores	Predictive-maintenance calendar / depot queue	dwell pager runs on weekly windows
Monthly	Performance review	Drift, FP/FN vs service outcomes, stale-truck audit	flag silent telemetry, not false-GREEN
Quarterly	Retrain if enough new labels	Champion / challenger promotion	trigger = new failures, not the date

Daily output by component:

Component	Daily inference output	Is it an RUL?
Clutch	RUL estimate + risk zone (anomaly-based, ~63-day lead)	Yes - RUL (caveated)
Alternator	Risk zone + maintenance window + GED=2 disturbance alarm	No - risk zone

Starter	Risk zone + maintenance window + persistent-RED dwell pager (weekly)	No - risk zone
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5. Retraining (quarterly)

Q8. Retraining compute - complete individual models, quarterly, on Azure (all 3 verticals)?

A. Retraining is quarterly only (champion/challenger, triggered by new confirmed failures - not the calendar). It is periodic retraining, not reinforcement learning. Cost depends on the model tier and is fleet-size-independent (the training matrix is small regardless of deployment size):

Model tier	GPU	Per quarterly run (3 verticals)	Per year (4 quarters)
Small models	A10 Spot	1-3 GPU-h (~Rs 94-Rs 329)	~Rs 450-1,300
Transformer (~500 veh/vertical)	A100 Spot	12-30 GPU-h (~Rs 1,260-Rs 2,867)	~Rs 5,000-11,500

Per vertical, the transformer quarterly retrain is ~Rs 1,700-3,800/year. Even on dedicated (PAYG) GPUs the all-vertical figure stays under ~Rs 60k/year. The real constraint is how many new labelled failures arrive, not the compute. Hours x rate in Annex A.3.

6. Expansion Strategy

Q9. Recommended expansion strategy?

A. Industrialize by component first - Clutch -> Alternator -> Starter -> Brake -> Battery - then extend across vehicle families. The data foundation and Truck Connect integration are built once and amortized, so each additional component is an incremental Rs 7 Lakhs; extending to other families is mainly re-calibration.

Q10. Single-Unit Truck clutch expansion?

A. ~10-12 weeks, Rs 10-12 Lakhs. Re-uses the proven clutch model and existing foundation; re-maps and re-calibrates features for the single-unit duty cycle, re-validates, and onboards into Truck Connect.

Q11. Mining / Off-Highway clutch expansion?

A. ~12-16 weeks, Rs 12-15 Lakhs. Off-highway duty cycles differ substantially (severe loads, low-speed/high-torque, harsh thermal/dust), needing more re-calibration, possible new failure modes and longer validation.

Q12. Alternator industrialization cost?

A. Rs 7 Lakhs incremental on the shared foundation: validated alternator model (AUROC 0.927) to production - GED-disturbance alarm, risk-window matrix, calibration, champion/challenger. Deliverable = risk score + maintenance window + GED alarm.

Q13. Starter industrialization cost?

A. Rs 7 Lakhs incremental: validated starter model (AUROC ~0.93) to production with its maintenance window and a persistent-RED dwell pager. Deliverable = risk zone + window + alarm, not a per-day countdown.

7. Commercial Model & Pricing

Q14. Recommended commercial model?

A. Offer both an Enterprise License (large committed / OEM-embedded fleets; capacity-based) and a Per-Vehicle SaaS subscription (variable/growing fleets; transparent Rs/vehicle/year). Both lets Daimler choose by deployment model and captures large-fleet commitments plus incremental adoption.

Q15. Recommended enterprise pricing?

Fleet size band	Enterprise price (full suite)
Up to 25,000 vehicles	Rs 1.5 Crore
25,000 - 50,000 vehicles	Rs 2.5 Crore
50,000 - 100,000 vehicles	Rs 3.5 Crore
Above 100,000 vehicles	Custom

Q16. Recommended per-vehicle pricing?

Component bundle	Price (per vehicle / year)
Clutch	Rs 150
Clutch + Starter	Rs 250
Clutch + Starter + Alternator	Rs 400
Full Suite	Rs 600

Pricing reflects avoided breakdowns and downtime; the underlying cloud compute is only ~Rs 15-23/vehicle/year (Section 3) - a small fraction of price.

8. What we need from Daimler

Q17. What information is still needed?

- **Vehicle variant mapping** - families/variants in scope and their clutch (and alternator/starter) hardware.
- **Deployment architecture** - Truck Connect integration contract (APIs, auth, latency/SLA); cloud / on-prem / hybrid.
- **Cloud preference** - Azure (assumed) and target region / data-residency (Central India assumed).
- **IT / security** - data residency, RBAC, audit logging, per-tenant isolation, signed-image / license controls.
- **Prediction frequency & retention** - once- vs twice-daily scoring (sets compute per Section 3) and prediction-history retention.

Annex A. Compute - hours x rate

Only GPU and CPU are hour-priced. Storage is per GB-month and platform/monitoring are flat fees - together about half the bill - so those are shown in their real units.

A.1 Rate card (Central India, 24-Jun-2026)

Resource	Unit	PAYG rate	Spot rate
A10 GPU (NV36ads_A10_v5)	\$/GPU-hr	\$4.48 (Rs 421)	\$0.828 (Rs 78)
A100 GPU (NC24ads_A100_v4)	\$/GPU-hr	\$5.142 (Rs 483)	\$0.95 (Rs 89)
D16ds_v5 CPU (16 vCPU - feature/ETL)	\$/instance-hr	\$0.918 (Rs 86)	~\$0.3 (Rs 28)
D8ds_v5 CPU (8 vCPU - inference)	\$/instance-hr	\$0.459 (Rs 43)	~\$0.15 (Rs 14)
Blob storage hot / cool / archive	\$/GB-month	\$0.020 / 0.010 / 0.002	-

A.2 Once-daily production @ 100,000 vehicles

Hour-based (compute), per month:

Job	Instance	Instance-hr/mo	= \$/mo PAYG	Spot option
Feature aggregation	D16ds_v5	435-1,307	\$400-1,200	\$130-390
Batch inference (x3 verticals)	D8ds_v5	218-654	\$100-300	\$33-98
GPU retrain (quarterly, amortized)	A100 Spot	~4-10 GPU-hr	-	\$4-10
Compute subtotal		~650-1,960 CPU-hr/mo	\$500-1,500	\$165-500

Not hour-based:

Line	Quantity	= \$/mo
Storage (raw 14-day hot + curated + features)	~12,000 GB-mo @ \$0.02/GB-mo	\$220-500
Registry + orchestration + monitoring	platform (flat) + per-run	\$200-520
Total once-daily @ 100k: ~\$950-2,560/mo = ~Rs 0.9-2.4 lakh; planning figure ~Rs 1.2-1.9 Lakh/month. Compute is model-agnostic - a transformer scores just as cheaply in daily batch (< 15 min/pass on CPU).		

A.3 Retraining (quarterly) - hours x rate

One quarterly retrain of all three verticals; transformer tier on A100 (small-model tier on A10):

Tier / GPU	Hours	= Spot / run	= PAYG / run	Per year (x4) Spot
Small / A10	1-3 GPU-h	Rs 94-Rs 329	Rs 442-Rs 1,354	Rs 450-1,300
Transformer / A100	12-30 GPU-h	Rs 1,260-Rs 2,867	Rs 5,988-Rs 14,692	Rs 5,000-11,500

- **Quarterly only:** 4 runs/year. GPU-hours/year = 48-120 (transformer) or 4-12 (small). Per vertical, the transformer is ~Rs 1,700-3,800/year.
- **Fleet-size-independent & checkpointed:** same cost for 25k or 100k vehicles; Spot eviction just resumes, so PAYG is only the rare must-finish ceiling.

A.4 Bottom line

Compute hours are nearly free: ~1-3 CPU-seconds per vehicle per day to score, and 48-120 GPU-hours per year to retrain all three transformers. About half the production bill is storage (GB-month) and flat platform fees, so the real cost levers are architecture (batch vs ~4x costlier always-on streaming) and retention - not the hours.

Annex B. Internal commercial notes

INTERNAL - NOT FOR EXTERNAL DISTRIBUTION. Remove before sharing with the customer.

B.1 Cost-to-price bridge

Consolidated 500-truck build (all components + data + Truck Connect + onboarding) ~ \$60.7-66.7k = ~Rs 57-63 Lakhs at Rs 94/USD. Rs 24 L clutch + Rs 7 L alternator + Rs 7 L starter = Rs 38 L covers the foundation + 3 verticals + Truck Connect readiness; the remaining ~Rs 19-25 L is brakes (greenfield), onboarding and the fuller Truck Connect integration. The offer is internally consistent.

B.2 Effort reconciliation (\$40/hr fully-loaded, Rs 94/USD)

Item	Price	~ eng-hours	~ eng-days
Rs 24 L clutch programme	Rs 24 L	~640	~80
Alternator / Starter (each)	Rs 7 L	~185	~23
Single-Unit Truck clutch	Rs 10-12 L	~265-320	~33-40
Mining / Off-Highway clutch	Rs 12-15 L	~320-400	~40-50

B.3 Points to confirm before quoting

- **1. Cost vs billing rate.** \$40/hr is fully-loaded (cost). Rs 24 L ~ 640 hours at that rate = near delivery cost; confirm whether \$40/hr is cost or billing and set the margin before quoting.
- **2. Enterprise vs per-vehicle.** Rs 1.5 Cr / 25,000 = Rs 600/veh = the full-suite per-vehicle rate. Decide annual vs perpetual; if perpetual, add an AMC (~18-22%/yr) for compute, support and retraining.
- **3. Starter vs alternator effort.** Starter is a 6-week plan phase vs alternator 4; both at Rs 7 L is fine if pilot reuse offsets it, else consider Rs 8-9 L for starter.
- **4. Architecture dependency.** The Rs 1.2-1.9 L/mo @ 100k assumes the batch architecture. Real-time / always-on serving moves it toward the heavier streaming profile - hence the prediction-frequency question.
- **5. Deliverable scope.** Clutch = RUL (~63-day lead). Alternator & starter = risk zone + window + alarm, NOT a per-day RUL. Do not sell daily RUL for alternator/starter.
- **6. Model tier is priced in.** The effort prices assume DL/transformer work; the transformer tier raises only compute (build + retrain), both within the quoted envelopes - not the engineering price.

B.4 Compute as % of price (margin sanity)

Enterprise band	License	Compute / yr (at top of band)	Compute %
Up to 25,000	Rs 1.5 Cr	Rs 6.5-10.3 L	4.3-6.9%
25,000-50,000	Rs 2.5 Cr	Rs 9.8-15.3 L	3.9-6.1%
50,000-100,000	Rs 3.5 Cr	Rs 14.6-22.8 L	4.2-6.5%

Per-vehicle: compute is ~Rs 15-41/veh/yr (fleet-dependent) against Rs 150-600 price -> ~3-7%. Healthy infra margin in both models; transformer retraining adds only ~Rs 5-12k/year fleet-wide, immaterial to the ratios.

Assumptions, sources & caveats

- **FX:** Rs 94 / USD.
- **Pricing snapshot:** 24-Jun-2026, Azure Central India, Linux PAYG / Spot - re-check the Azure calculator before commitment.
- **Architecture:** batch-first (daily CPU batch scoring, on-demand Spot GPU for quarterly retraining); not always-on streaming.
- **Model tiers:** small models (A10) for small data; transformer (A100) at ~500 veh/vertical. Production inference is daily CPU batch in both - run-rate is model-agnostic.
- **Deployment / training scale:** deployed ~100,000 vehicles; trained ~500 vehicles/vertical.
- **Scope of compute figures:** cloud compute + data platform only. Excludes one-time build, dashboard hosting, support, Truck Connect integration engineering and data egress.
- **Sources:** DICV 500-Truck Scale-Up Plan (Consolidated) and the Azure Deployment / Retraining / GPU plans (v2 batch deployment supersedes v1).
- **Status:** planning estimates, NOT a quotation.

DICV BharatBenz Predictive Maintenance - Truck Connect Predictive Intelligence - Commercial Q&A - 2026-06-26